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Development of an eco-ideation tool to identify stepwise greenhouse gas emissions reduction options for consumer goods

N.M.P. Bocken [a], J.M. Allwood [a,*], A.R. Willey [b], J.M.H. King [b]

[a] Department of Engineering, University of Cambridge, Cambridge CB2 1PZ, UK
[b] Unilever Safety & Environmental Assurance Centre, Bedford MK44 1LQ, UK

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ABSTRACT

Pressure on consumer goods manufacturers to develop new products with significantly less environmental impact is growing, through increased consumer awareness of environmental issues and governments setting ambitious emissions reductions targets. A strategic response to this pressure is to prepare a portfolio of innovative product ideas to meet a range of future emissions reductions targets. However, although extensive work exists on ideation (the generation of novel product ideas) and eco-design (design for reduced environmental impact), eco-ideation (generation of ideas that particularly aim to reduce environmental impacts) has had little attention. The challenge of eco-ideation is to release the creativity of a broad range of employees, only few of whom may be familiar with the drivers of environmental impact.

This paper proposes a novel tool to facilitate the generation of radical product and process ideas giving step-change reductions in greenhouse gas emissions. The features of products and processes that drive greenhouse gas emissions across the product life cycle were characterized with a set of indicators. A simple visual tool was created to show these indicators on a sliding scale between best and worst imaginable performance. A leading question associated with each slider was designed to stimulate lower impact ideas. The tool was iteratively refined and simplified through structured testing with individuals from across a range of roles and differing knowledge of environmental impacts. The final eco-ideation tool used 14 scales, with leading questions for each scale developed to stimulate idea generation.

The tool's effectiveness was evaluated through use in a series of 15 individual workshops and compared with the outcomes of an equal number of conventional 12-person group-brainstorming sessions. The comparison suggests that using the simple tool generally leads to a wider range of more radical ideas than emerge from group brainstorming.

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1. Introduction

Consumer goods manufacturers face increasing pressure from governmental bodies, environmental pressure groups, retailers and final consumers to reduce the environmental impacts of their products. In the UK, for example, the number of consumers buying locally produced food has nearly doubled between 2006 and 2009 (DEFRA, 2009). There is a growing consensus that the threat of global warming is severe, and in response, rapid stepwise reductions in greenhouse gas emissions are required (IPCC, 2007). Consumer goods contribute to anthropogenic global warming through emissions arising from raw material sourcing, distribution, manufacture, storage, use and disposal.

Unilever is a large multinational fast moving consumer goods manufacturer, so called because its products need frequent replenishment after a limited number of uses. To indicate the company's size, Unilever purchases approximately 12% of the world's black tea, 6% of its tomatoes and 3% of its palm oil (Unilever, 2009). Unilever commissioned this research to identify a range of innovative options to service the same need as existing products, but with significant reductions in associated greenhouse gas emissions.

The paper, therefore, describes the development of an 'eco-ideation' tool. Ideation is the process of generating ideas, or according to Osborn (1957, p. 115) of 'piling up alternatives.' Eco-ideation, as defined here, is the generation of ideas that reduce environmental impacts (in this case greenhouse gas emissions) throughout the product life cycle of consumer goods. The ideas that enable stepwise reductions are expected to be more *radical*, in contrast with the *incremental* ideas driving conventional product development. Radical innovation has also been called

* Corresponding author. Tel.: +44 1223 338181; fax: +44 1223 332662.
E-mail address: jma42@cam.ac.uk (J.M. Allwood).

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discontinuous or, according to Bower and Christensen (1995), disruptive. Govindarajan and Kopalle (2006) describe *radicalness* as the characteristic of innovations requiring substantially new technology relative to current practice and *disruptiveness* as the characteristic of innovations with limited initial appeal before attracting mainstream consumers and thus displacing existing products. By these definitions, the most radical and disruptive innovative product ideas may initially be less attractive for companies, which nurture consistent brand values to serve a mass market.

However, if such companies have an ambition to implement step-change reductions in the impacts of their products, radical and disruptive ideas may be required, and having a portfolio of options for future change allows pre-emptive evaluation of consumer response and opportunities.

2. Ideation and eco-design in the literature

Although eco-ideation has not been widely explored, development of a tool in the area can build on substantial prior work in the separate areas of ideation and eco-design. This section examines the literature in both areas and reviews the few prior attempts to apply structured idea generation specifically to achieve improved environmental performance.

2.1. Group and individual ideation

Is creativity a collective or individual activity? Perhaps surprisingly, although in the arts there are few if any examples of masterpieces created by groups, most corporate attempts to generate ideas are based on collective work. Individual idea generation tools such as morphological analysis and TRIZ have been well documented, but the most common tools are either group based or require multiple iterations with many experts.

The most popular ideation methods used in companies involve many people and three well known examples dominate: focus groups, technology roadmapping and Delphi studies. *Focus groups* typically involve 8–12 people, facilitated by a moderator, exploring opinions and attitudes and testing new product ideas. Where these groups aim to create new ideas, their action is referred to as 'brainstorming' which aims to generate but not evaluate a stream of new concepts. *Technology roadmapping*, initially developed by Motorola, is a planning process in which groups aim to prioritize products and technologies for future development. Phaal (2004) found that the technology roadmap itself may look simple, but the process of developing it is complex, involving many experts. The process of roadmapping can be rewarding, but several iterations may be required before the full benefits of the approach are achieved. *The Delphi study* is a method to retrieve and organize a wide range of expert opinions on some phenomena, such as the emergence of a new technology. The technique requires two responses from each individual participant: firstly to identify a range of opinions or options; secondly to prioritize the complete set generated in the first round. Schmidt (1997), and Okoli and Pawlowksi (2004) building on Schmidt's approach, have proposed systematic methods to conduct a Delphi study, to improve data collection and reporting of results. Powell (2003) having analyzed various studies on the Delphi technique found that the technique systematically includes expert opinions, but that it suffers a methodological weakness due to lack of clarity on how consensus is built. These three group ideation methods are all iterative, with experts attending one or more sessions to generate and prioritize ideas.

Evidence about the quantity and quality of ideas generated suggests that, although popular, group ideation sessions are not always effective. Taylor et al. (1958) investigated whether group participation in brainstorming facilitates or inhibits creative thinking. They compared the outcomes of brainstorming with 12 four-person groups to those from 48 individual brainstorms. After the 48 individual sessions, the ideas were randomly clustered in 12 groups of 4 individuals each, to create "nominal" groups to compare against the real 4 person groups. The experiment showed that nominal groups generated more ideas than real groups. Taylor et al. (1958) therefore concluded that group participation when brainstorming inhibits creative thinking. Experiments by Fern (1982) also showed that despite the popularity and assumed effectiveness of focus groups, individuals usually generate more and higher quality ideas. Nijstad et al. (2006) claim that 50 years of research has demonstrated that groups generate fewer ideas than individuals. Reasons for the apparent ineffectiveness of group brainstorming presented by Sutton and Hargadon (1996) include production blocking (listening to others blocking thinking), social loafing (individuals choosing not to participate), free riding (individuals in groups feeling less responsible and accountable) and to a lesser extent evaluation apprehension (not sharing ideas for fear of the reaction of others).

Some of the problems with group brainstorming may be mitigated through avoiding group contact, for instance by surveying participants individually (as happens in Delphi studies), or through electronic brainstorming in which group sessions are mediated by computers (Gallupe et al., 1991). According to Dennis and Valacich (1999), electronic brainstorming is more effective with larger groups. Production blocking for example may not occur, because the opinions or ideas of others do not interfere with participants' creative thinking. However, despite the advantages of computer mediated brainstorming, conventional face-to-face group brainstorming remains popular. Stroebe et al. (1992) suggest that participants perceive face-to-face group sessions to be effective, perhaps because of low failure rates (instances when no ideas are generated) or because they view their own performance too positively. Paulus et al. (1993) found that participants generally believed that groups generate more ideas than individuals and showed that individuals in brainstorming groups disproportionately claim more of the resulting ideas as their own. Nijstad et al. (2006) call this the 'illusion of group productivity.' So, although group brainstorming may feel effective, this is not always the case.

In contrast to these group methods for ideation, Osborn (1957, Preface) discusses a list of structured methods that could be used to generate ideas *individually* including attribute listing, forced relationships, and morphological analysis. These methods seek to restructure a problem to facilitate creative solutions through listing all the separate elements of a product and rethinking these (attribute listing), through forcing relationships between previously distinct products (forced relationships) or through listing all parameters that define a product or technology in a structured format (morphological analysis). Morphological analysis developed by Zwicky (1969) requires the idea generator to divide the problem into different parameters; often in a 3-dimensional grid (morphological box) and to consider all possible combinations of parameter settings as potential solutions. The theory of inventive problems, TRIZ, (as described by Altshuller, 1996) is based on a list of 40 design principles to address contradictions in design (such as a desirable decrease in weight leading to an undesirable loss of strength). Although some simplified methods have been developed such as TRIZ system-operator or nine-window TRIZ (illustrated with examples in Mann, 2001) mastery of individual ideation methods still requires skill and time, because of the number of principles involved and the need to identify and model all potential variables.

The evidence presented here appears to suggest that group ideation is generally less effective than individual creativity, but that existing individual tools may be difficult to use. This suggests that a useful eco-ideation tool should be individually operated, but should not demand significant training for each new user.

2.2. Eco-design tools

Eco-design, the design of products with reduced environmental impacts, dates from the 1970s, when environmental pressure groups such as Greenpeace and institutions such as the US Environmental Protection Agency were formed. An early book on eco-design, 'Design for the real world' (Papanek, 1985) aimed to help designers to envisage the wider environmental impact of products, but Baumann et al. (2002) in their review paper found that the area of eco-design really became established in the 1990s. Synonyms for eco-design include environmentally conscious design, green design, design for *X* (where *X* might be low material use or environmental impact), sustainable design (incorporating wider social and economic impacts as well as environmental concerns) and dematerialization. Interest in the area has led to development of a large number of eco-design tools which and, based on the classifications used in reviews by Baumann et al. (2002) and Byggeth and Hochschorner (2006), can be classified broadly as guideline, evaluative, comparative and trade-off tools.

Guideline/checklist tools (terms used by Baumann et al., 2002) or equivalently prescribing tools (the term used by Byggeth and Hochschorner, 2006) provide the user with a set of rules that help to improve the environmental performance of products and processes. Luttrop and Lagerstedt's (2006) '*Ten golden rules*' for instance are developed as a 'pedagogic summary' of guidelines from companies such as Volvo and describe in detail areas such as more efficient material use, simplicity in design, and choosing low impact materials. The *Sustainable Product and Service Development* (*SPSD*) checklist by Maxwell and Van der Vorst (2003) provides pragmatic guidelines including recommendations on materials, production, distribution, consumption, end of life, and optimized functionality.

Evaluative tools support the user in the improvement of the environmental performance of product and process design. Examples include the Environmentally Responsible Product Assessment Matrix (ERPA) (Graedel, 1997) and the Materials Energy & Toxicity (MET) Matrix (Van Berkel et al., 1997) which aim at a simplified evaluation of the environmental performance of a product in order to define areas for improvement.

Comparative tools facilitate comparison of the environmental performance of different products, sometimes including business trade-offs such as costs. The tools often have the shape of a spider web allowing multiple products to be compared against different criteria. Examples include the Life-cycle Design Strategy (LiDS) wheel, of which later versions include the Eco-compass by Fussler and James (1996) and the eco-design strategy wheel by Van Hemel and Cramer (2002) based on Van Hemel and Brezet (1997). The Eco-compass developed at chemicals company Dow includes the factors service extension (similar to optimizing product life in the LiDS wheel), revalorization (developing more durable, recyclable goods and reducing material intensity), energy intensity, mass intensity, health and environmental risk, and resource conservation. Van Hemel and Cramer (2002) compared 33 possible eco-design solutions in an 'eco-design strategy wheel'. The eco-design wheel includes seven "normal" strategies: selection of low impact raw materials; reduced material use; optimized production; optimized distribution; reduced impact during use; optimized initial lifetime; management of end-of-life options. A separate category, 'New Concept Development', includes more innovative options such as dematerialization, shared use, and functional optimization. The eco-design solutions list described by Van Hemel and Cramer (2002) is extensive and many other checklists in the literature include a subset of them.

Other eco-design tools aim to show a clear *trade-off* between business and environmental factors. Byggeth and Hochschorner (2006) present an overview of fifteen different trade-off tools. A corporate example of such an approach is BASF's eco-efficiency analysis, comparing life cycle environmental impacts against life cycle costs (Saling et al., 2002). Contrasting life cycle costs and life cycle impacts allows for the identification of unsustainable life cycle phases (e.g. high cost and high environmental impact).

Sarkis (2003), Vermeulen and Ras (2006) and Linton et al. (2007) describe strategies aiming to create *supply chains* with reduced environmental impacts. Both Vermeulen and Ras (2006) and Linton et al. (2007) emphasize the elimination and reduction of waste, energy and by-products along the supply chain. Sarkis (2003) proposes an Analytical Network Process methodology to support decision making for green supply chain management with strategies to reduce the impacts of procurement, packaging, production, distribution, and reverse logistics. Strategies recommended for procurement, for instance, include the purchase of recyclable, reusable and recycled materials, and the selection of vendors (for example, vendors with ISO 14000 certification may be preferred). This approach however requires significant user input and may involve complex modeling. Linton et al. (2007) and Vermeulen and Ras (2006) emphasize the importance of reducing environmental impacts throughout the supply chain, including some recommendations on how to do this, but do not provide a tool or checklist.

The literature of eco-design methods and tools is thus extensive but most approaches assume that a set of options exist so must be evaluated, compared or traded-off across different criteria.

2.3. Eco-ideation

The tools identified in Section 2.2 were not developed for eco-ideation, but were generally designed for more evaluative purposes, such as comparing business against environmental objectives. This early evaluation, which can be termed 'judging' (Osborn's terminology, 1957, p. xix Preface) may inhibit creative thinking and prevent the pursuit of good ideas. However, in practice some evaluation tools may be used to assist with eco-ideation: Charter and Tischner (2001, pp. 269–281) provide an overview of eco-design tools (Eco-compass, eco-design checklists) which could assist group brainstorming, and Jones et al. (2001) specifically propose the use of Standard Design Process Form (SDPF) and Product Ideas Tree (PIT) diagrams in combination with eco-design tools to improve structuring eco-design-focused brainstorms.

Few methods have been developed for the specific purpose of eco-ideation. Byggeth and Hochschorner (2006) view the morphological box (Zwicky, 1969) as a potential *eco-idea generation tool* but the approach was not specifically developed for eco-design. In addition, a body of work has developed applying TRIZ to eco-design, including Chang and Chen (2004) who developed CAD software integrating TRIZ and eco-innovation, Kobayashi (2006) who proposed a Life Cycle Planning approach building on TRIZ factors (weight, time, temperature) and Low et al. (2000) who explored the application of TRlZ parameters to services such as car sharing. Strategies identified in the literature where TRIZ is used for eco-design include: increasing service intensity; replacing products with services; reducing energy intensity; decreasing weight and volume; increasing useful life span.

Nissen (1995) developed 'the ideal eco-product-approach', a stepwise idea generation approach for eco-ideation in which the user initially describes the functions of the product and potential technology options, then estimates the environmental impact of these options, develops extreme designs to reduce these impacts (an approach also proposed by Osborn, 1957), then combines features of the extreme ideas to form a new product concept. Nissen's approach provides the closest prototype for the need addressed in this paper. However, it assumes that the user already understands product environmental impacts (knowledge users



Fig. 1. Research methodology.

may not have), the approach includes a commercial evaluation which may inhibit creativity, and no reports have been published on the effectiveness of the approach.

Thus, the motivation to develop a new eco-ideation approach is that there are currently no easy to use eco-design tools that can be used by a broad range of actors across different functions to generate eco-innovative ideas. The eco-ideation tool is developed for a wide range of actors as novel product and process ideas originate from different functional areas (product development, science, marketing). In addition, the evidence of Section 2.1 suggests that popular group ideation techniques may not be as creative as anticipated, firstly because of production blocking and secondly because early evaluation will tend to eliminate more radical ideas. Finally, ideation sessions must be guided by appropriate awareness of the drivers of environmental impacts which may not be well understood across all corporate functions.

3. Tool development

A new eco-ideation tool is developed based on a set of key indicators that predict greenhouse gas emissions through the product life cycle. The tool comprises a set of visual 'sliders' to show each indicator setting varying between maximum and minimum influence on greenhouse gas emissions. Each slider is associated with one or more leading questions to stimulate creative product and process ideation by contrasting 'extremes' (e.g. zero waste, compared to 100% waste at end of life).

A stepwise approach, illustrated in Fig. 1, was applied to develop, test and design the new tool.

Moving across the figure from left to right, the identification of an initial set of parameters will be discussed in Sections 3.1, from which an initial tool design is created in Section 3.2. The initial tool was tested in pilot studies described in Section 3.3 and then iteratively refined in a series of structured individual workshops described in Section 3.4. The resulting final tool is presented in Section 3.5. The effectiveness of the tool will be investigated in the next section by comparing the quality and quantity of ideas stimulated by use of the tool with the outcomes of group-brainstorming sessions.

3.1. Identifying indicators

Two sources were used to identify an initial set of key indicators of emissions associated with a product. Firstly, discussions were held with a broad range of experts within Unilever to identify common metrics used to characterize how products are made, used and sold. In addition, the company has extensive experience in the use and analysis of life cycle assessments for a wide range of food and home and personal care products, and these were examined for key measures. A combination of these sources led to the development of the list of indicators in Table 1.

Secondly, the literature on eco-design reviewed in Section 2.2 was investigated to create a catalog of indicators used in existing checklists and tools. Indicators were selected if relevant to energy use or greenhouse gas emissions and applicable to consumer goods. This search led to the list of indicators in Table 2.

These two tables of existing indicators have some overlap, but provided a base set for initial tool development.

3.2. Initial tool

Tables 1 and 2 were used to develop an initial set of 36 indicators of greenhouse gas emissions, with around 8–10 per product life cycle phase: sourcing; manufacturing; distribution, warehousing

Table 1
Factors identified from company's product and process analysis.

Area of concern	Indicator	Evaluation/Quantification
CO_2 emissions from the	*Distance*	
distance traveled from	Sourcing, production, retail, consumer	Distance traveled
raw material sourcing	Mode of transport	Efficiency and emissions of mode of transport
to production, retail and	Load mode of transport	% Full (area, weight, volume)
final consumers.	(area, volume, weight)	
CO_2 emissions resulting from	*Pack*	
the product pack design. Is	Stock keeping unit (SKU)	Efficiency of packaging design: e.g. individual wrappings or bulk.
the design optimal for its purpose?	Re-packaging	Number of times product is repacked throughout supply chain
	How product comes out of pack	E.g. Direct application, propellant, battery-powered
	Ingredient (or product) – to-pack ratio	% Ingredients to packaging – verify fit for purpose and overdesign
CO_2 emissions embodied in raw	*Components*	
materials purchased, and recyclability.	Type of materials and components	CO_2 emissions embodied in raw materials (from extraction, processing)
	Number of materials and components	How easily can product be recycled based on material mix?
CO_2 emissions resulting from "bulkiness"	*Dimensions*	
of the product (product, transport,	Weight	Weight/distance traveled. CO_2 emissions embodied in raw materials
warehousing, retail)	Volume	Space consumed in warehousing, retail, and consumer homes
	Density (compact or outsized) and	Level of concentration or compaction of product. Can it be
	Phase (e.g. solid, liquid, vapor)	denser and less diluted?
CO_2 emissions from temperature changes	*Temperature changes*	
(heating and cooling) throughout supply	Sourcing and production	Temperature changes (heating, cooling) for storage and processing
chain, consumer use and disposal	Retail	Temperature for storage (hot or cool) and transport
	Logistics/Warehousing	
	Consumer use	Storage (e.g. freezer) and product usage (e.g. boiling)
	Disposal/end of life	E.g. Incineration of waste; melting for recycling
CO_2 emissions due to frequent replenishment,	*Durability and end of life*	
and lack of reusing products and materials	Performance over time	Useful product life. Is it durable and does it keep performing well over time?
	Shelf/stock life	Waste due to surpassing product lifetime. For durable goods: esthetic
	Best by/use by time	value over time
	Reusability & Recyclability	Efficiency of material use. Can the product be reused or recycled?

Table 2
Indicators of emissions impacts from the literature.

Indicators	Literature sources
Weight, Volume, Dimensions	Sustainable product analysis (SPA) checklist;
Compaction	Byggeth et al. (2007)
Light rather than heavy	Ten Golden rules; Luttrop and Lagersted (2006)
Ingredient-to-pack ratio	TRIZ & eco-design; Kobayashi (2006)
	TRIZ & eco-design; Chang and Chen (2004)
	Sustainable Product and Service Development (SPSD) checklist; Maxwell and Van der Vorst (2003)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)
Reduce total product use	SPA checklist Byggeth et al. (2007)
Alternative ways to provide service	TRIZ & eco-design, Chang and Chen (2004)
Shared use	SPSD checklist; Maxwell and Van der Vorst (2003)
Integrating and extending functions	Eco-design strategy wheel; Van Hemel and Cramer (2002)
	TRIZ factors & eco-design; Low et al. (2000)
Supply chain wide indicators	SPA checklist; Byggeth et al. (2007)
Reduce need for transport	Sustainable supply chains;
Reverse logistics (e.g. reuse)	Linton et al. (2007)
Temperature & Time	TRIZ & eco-design; Kobayashi (2006)
Energy and material reduction	Green supply chains; Vermeulen and Ras (2006)
	TRIZ & eco-design; Chang and Chen (2004)
	SPSD checklist; Maxwell and Van der Vorst (2003)
	Green supply chain management; Sarkis (2003)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)
	TRIZ factors & eco-design; Low et al. (2000)
Elimination and reduction of:	SPA checklist; Byggeth et al. (2007)
Waste from product + pack	Sustainable supply chains; Linton et al. (2007)
Energy consumption	Ten Golden rules; Luttrop and Lagersted (2006)
Manufacturing + use by-products	TRIZ & eco-design; Kobayashi (2006)
	Strategies for environmental impact reduction; Handfield et al. (2005)
	TRIZ & eco-design; Chang and Chen (2004)
	SPSD checklist; Maxwell and Van der Vorst (2003)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)
Substitution of raw materials	SPA checklist; Byggeth et al. (2007)
Substitute, Reduce, Recycle	Ten Golden rules;
Investing in better materials	Luttrop and Lagersted (2006)
Simplicity	TRIZ & eco-design; Kobayashi (2006)
	Strategies for environmental impact reduction; Handfield et al. (2005)
	TRIZ & eco-design; Chang and Chen (2004)
	SPSD checklist; Maxwell and Van der Vorst (2003)
	Green supply chain management; Sarkis (2003)
	TRIZ factors & eco-design; Low et al. (2000)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)
Consumption/use of the product	SPA checklist; Byggeth et al. (2007)
Energy required in use phase	Sustainable supply chains; Linton et al. (2007)
	Ten Golden rules; Luttrop and Lagersted (2006)
	TRIZ & eco-design; Chang and Chen (2004)
	SPSD checklist; Maxwell and Van der Vorst (2003)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)
Life time of the product	SPA checklist; Byggeth et al. (2007)
Long technical + esthetic lifetime	Sustainable supply chains; Linton et al. (2007)
Reusability and recyclability	Ten Golden rules; Luttrop and Lagersted (2006)
Durability, reparability, upgradability, preciousness	TRIZ & eco-design; Kobayashi (2006)
	Strategies for environmental impact reduction; Handfield et al. (2005)
	TRIZ & eco-design; Chang and Chen (2004)
	SPSD checklist; Maxwell and Van der Vorst (2003)
	Eco-design strategy wheel; Van Hemel and Cramer (2002)

and retail; use and disposal. The indicators were illustrated on sliders, and leading questions were designed to prompt thinking on how the sliders could be moved in the direction of lower emissions. The questions were developed to stimulate tool-users to rethink delivery of a certain product value or function, for example: can the product function be delivered with a compact and light product, or use fewer temperature cycles? A typical question about end-of-life might be 'is the product reusable?' The answer leading to lowest emissions might be an 'endlessly reusable' product and the worst option could be a 'disposable and non-recyclable' product. Other questions included 'must the product be stored at ambient, chilled or frozen temperature?' and 'how many active ingredients are typically left in the pack after use?'

Fig. 2 shows the slider for 'transport loading', which indicates whether the capacity of the mode of transport (including the return journey) is fully used, and aims to question whether the product and process design enables full loads. The best option is, of course, transport with a 100% full load.

The aim of each leading question is to stimulate a separate 'mini-ideation' session for the tool user.

3.3. Pilot test

A pilot test was conducted to ensure that the scales correctly reflected low and high greenhouse gas assessments and to eliminate redundant questions: if the outcomes (ideas) from the mini-ideation sessions were similar, the separate questions were merged. For example, two questions which were merged were 'is the product solid, powder, gel, liquid or vapor?' and 'how diluted is the solution?' In the updated tool this question was changed to 'what does the active solution look like, and can it be less diluted?'

In the pilot test it was found that some repetitive questions could be merged, by changing the overall categories of questions. The initial tool for example included questions about 'location' at multiple life cycle stages (sourcing, manufacture). Therefore, instead of including questions about location of suppliers and manufacturing separately, one question was included about 'location' in the updated version: 'what is the distance from raw materials to production, retail and final consumer use, and how can this be minimized?' This process reduced the number of categories in the tool from four to two: Product design, and Processing & Supply chain. After merging the questions and changing the overall question categories, the total number of sliders was reduced from 36 to 15.

3.4. Individual workshops

Fifteen individual workshops were organized to test and iteratively improve the initial tool's questions and sliders and its ease of use. The sessions were organized to generate ideas for different types of products relevant to Unilever. Although the majority of the sessions were single person sessions facilitated by one of the authors, three out of the fifteen sessions were with two-people. Different people attended the sessions, including experts from the areas of environmental sustainability, new product development, product and process engineering, novel material processing, novel packaging development, and global supply chain processes.

During the individual workshops, participants were asked to generate as many radical ideas as possible using the tool. This is in



Fig. 2. Example slider for 'Transport loading'.

PRODUCT DESIGN	1 **DELIVERY OF VALUE:** Can value (e.g. joy, confidence) and functions be delivered without a product?	Delivering value/ functions without a product	Shared use/ Reduced product need	Product (not shared)
	2 **PRODUCT FUNCTIONALITY:** Does the product serve several functions, or can the function be integrated in something else? Are complementary products required to make it work?	Multi functional / No complementary tool required	Single function or tool required	Single function / Complementary tool required
	3 **COMPACTING:** How compact is the product? Can the product be sized up or diluted at a later stage of the product life cycle?	Compact / Dense	Medium compactness and density	Outsized / Dilute
	4 **COMPONENTS/ INGREDIENTS:** How energy intensive are components and ingredients? For example, plastic is less energy intensive than steel; and paper is less energy intensive than plastic; Animal-based is more energy intensive than vegetable based, keeping weight and distance travelled equal.	Non-energy-intensive	Room for improvement	Energy-intensive
	5 **RAW MATERIAL EFFICIENCY:** How efficiently are raw materials used?	Efficient: Low use and loss. Reusable, Recyclable	Room for improvement	Inefficient: High use/loss. Disposable, Non-Recyclable
	6 **DURABILITY/ EFFECT/ EXPERIENCE:** Is the product durable? Is the effect or experience long-lasting? Can the product be used in a targetted way to fulfill the required functions?	Durable / Long-lasting / Can be used in a targetted way	Medium effect, not very targetted	Short-lifetime / Short-lived effect / Unfocused
	7 **END OF LIFE:** Can the product be entirely used up, or is it endlessly reusable?	0% waste at end of life / Endlessly reusable	Some recycle, repair, disassembly, reuse issues	100% waste / Not reusable, repairable or recyclable
PROCESSING & VALUE CHAIN	8 **INTERMEDIATE PACKAGING:** How often is the product repacked throughout the supply chain?	One shelf ready pack	2-3 repackings	4 or more repackings
	9 **DISTANCE:** What is the distance travelled from raw materials sourcing to production, warehouses, retail and final consumer use? Can some processing be done at home or locally?	Local/ Home	National/ Regional	Global
	10 **LOAD TRANSPORT:** Is the capacity of the mode of transport fully used (also on the return journey)? Does product and process design enable full loads?	100% full	50% full	0% full
	11 **TEMPERATURE TRANSPORT:** How much non-ambient temperature transport is required?	None	Some	Much
	12 **TEMPERATURE CYCLES:** How many temperature changes take place throughout the product life cycle including use? E.g. boiling, freezing and cooling.	None	Some	Many
	13 **TEMPERATURE FOR STORAGE:** How much non-ambient temperature storage of raw materials and final products is required throughout the product life cycle including use?	None	Some	Much
	14 **STORAGE TIME:** How long is the final product stored throughout the whole supply chain? Can it for example be made more to demand?	Zero days	1 day - 6 months	More than 6 months

Fig. 3. Eco-ideation tool.

line with the rules codified by Osborn (1957) to improve the effectiveness of brainstorming: aiming for quantity, not quality, and not judging ideas as this may inhibit creativity. The individual workshops lasted for around 1 h. The participants in the first three workshops were experts in low energy processing and practitioners in the LCA field. These sessions were especially focused on verifying whether the sliders (such as the example in Fig. 2) correctly reflected likely greenhouse gas emissions. Later participants included process, supply chain and new product development experts. Most participants did not have specific expertise about the product that was the subject of their session.

In order to minimize the number of failures (non-responses) the facilitator (each time the same person) sometimes probed the individual by giving an example of an idea from a different product area. Shalley et al. (2004) for instance suggest that in the supervisor-subordinate context more encouragement led to more ideas, so probing is expected to lead to a higher response rate. Based on the examples used for probing, the questions and sliders

were refined and the tool was gradually improved. The tool was continually updated and the types of changes made at each point can be classified as: the addition of clarifying examples, rephrasing questions, and changing the order of questions.

To illustrate how *clarifying examples* were added, two examples are given. During the first workshop with two environmental sustainability experts, an example was added to a question about the energy intensity of (pack) components: 'Plastic is less energy intensive than steel; and paper is less energy intensive than plastic'. After the second workshop with two LCA experts, a clarifying example was added about energy intensity of ingredients: 'Animal-based ingredients are more energy intensive than vegetable-based ingredients'. These examples were intended to help the tool-user without expertise in emissions calculations to generate better-informed ideas.

The *order of questions* was also changed, so that earlier questions generally led to the most answers. For example,1. 'Can value (e.g. joy, confidence) be delivered without a product?' and 2. 'Does the product serve several functions, or can the function be integrated in something else? Are complementary products required to make it work?' The questions people found to be more difficult to answer, for instance about components and ingredients, were moved more to the middle of the list, and surrounded by questions perceived to be easier.

Some questions were *rephrased* to generate a wider response of answers. One question that significantly changed was: 'What is the product's lifetime?' which in the final version was: 'Is the product durable? Is the effect or experience long-lasting? Can the product be used in a targeted way to fulfill the required functions?' This question was rephrased to better reflect the broader concepts of 'durability' and 'product performance' than the more limited concept of just product lifetime.

Providing clarifying examples, rephrasing questions and changing the question order helped to refine and improve the tool. During the last five sessions, the questions were not changed further, and only minor adjustments to the order were made.

3.5. Final tool

The final tool consists of fourteen questions split into seven questions on 'product design' and seven for 'processing & value chain'. The tool questions and sliders are shown in Fig. 3.

4. Tool evaluation

The quantity and quality of ideas generated are typical measures used to assess the effectiveness of ideation (Sutton and Hargadon, 1996). This approach was used to compare trials of the new tool used by individuals against the results of face-to-face group-brainstorming sessions in the same company. The group-brainstorming sessions involved multidisciplinary groups, including experts from R&D, marketing, LCA and others, and had been organized to discuss current ideas in the innovation pipeline intended to reduce greenhouse gas emissions and to generate new ideas to reduce the footprint further. The brainstorming group sessions involved around 12 people each.

The quantitative assessment of the two approaches was made by comparing the number of non-overlapping ideas generated in the individual workshops using the new tool, with those generated during face-to-face group brainstorming. The qualitative analysis examined the ideas generated by the two approaches to assess how radical they were. Both measures are clearly subjective, as must be any assessment of ideation prior to eventual new product introduction.

The quantitative comparison is shown in Table 3. 70% of the individual sessions generated more ideas than group sessions.

Qualitatively, there were some similarities in the outcomes of the group and individual tool sessions. Both types of sessions resulted in ideas on concentrated product options (e.g. an ultra-concentrated household care product to be diluted with water at home), products that require less heated water in the use phase, and reusable products (a more durable and esthetic reusable pack). The groups generated a variety of ideas, but compared to the individual sessions there were more ideas on establishing collaborations with external parties to improve wider life cycle impacts (e.g. retail and suppliers) than ideas where the product design significantly changed. The individuals generated more ideas on radically changing product design, to deliver the same product value in a different way. Examples include 10-in-1 product functionality and embedding fast moving consumer good functionality in more durable goods to reduce total demand for products. Other ideas could disrupt the whole supply chain layout by moving more 'work' to the consumer (e.g. refillable product models). These options require technological changes, and may (initially) not appeal to mainstream consumers, which is typical for more disruptive, radical innovations (Govindarajan and Kopalle, 2006). To give an example, Fig. 4 shows the ideas generated by the group brainstorm and individual sessions for product B, classified by five themes, which emerged from the brainstorming sessions. "Alternative delivery of value" refers to question 1 from the eco-ideation tool and is about delivering the same product function in completely different ways. Although the groups considered more ideas on partnerships (supply chain), the individuals considered more options on product design and alternative ways of delivering the same product.

In some cases, the ideas generated during individual workshops were more extreme versions of the ones generated by the groups, suggesting more radical changes to the product, but potentially leading to greater reductions in greenhouse gas emissions. An example of a radical idea from one of the individual workshops is a product that allows hair washing once per month or year (as opposed to several times per week). This would be likely to reduce total product impacts, unless extreme chemicals were required.

5. Discussion

The goal of the research reported in this paper was to provide a consumer goods manufacturer with an approach to develop a wide range of ideas to make stepwise reductions in the greenhouse gas emissions associated with its products. Such ideas should be long term and should not logically follow from business as usual.

Table 3
Quantitative comparison.

	Product	No. of ideas individual session	No. of ideas group session
1	Product A	16	15 (but 5 discarded)
2	Product A	11	
3	Product B	16	14
4	Product B	15	
5	Product C	14	16 (but 6 discarded)
6	Product C	19	
7	Product D	25	15
8	Product D	18	
9	Product E	15	15
10	Product E	16	
11	Product F	16	8
12	Product G	13	14
13	Product G	28	
14	Product H	20	11
15	Product I	14	10

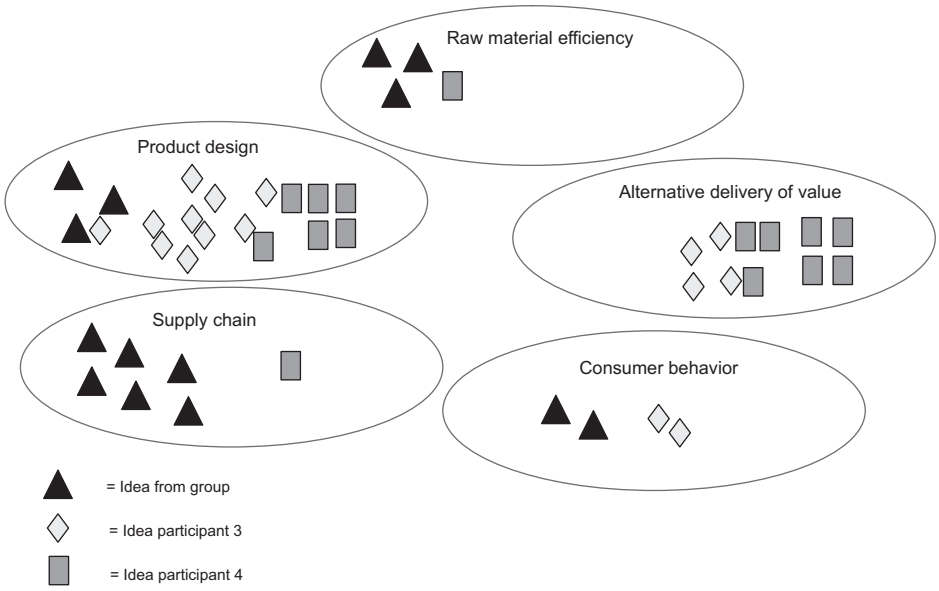


Fig. 4. Comparison of types of ideas generated for product B. Note: Closely linked ideas appear near to each other.

The paper has described development of an eco-ideation tool to meet this goal, and the evidence of the previous section suggests that the tool leads to creation of a wider range of more radical ideas than group brainstorming, as anticipated in the literature reviewed in Section 2.

Osborn (1957) argued that the generation of large quantities of ideas should be encouraged and people should build on, rather than criticize each other's ideas. When there is an encouraging open atmosphere people may feel happier to share ideas, so constructive probing which took place during the individual sessions may have been an important factor as the brainstorm sessions were facilitated by different people with varying facilitation styles. In addition, the level of creativity differs per person, which King et al. (1996) for instance explained through the five-factor model of personality. Personal motivation and interest will have affected outcomes of the sessions too. The individuals generating ideas by use of the eco-ideation tool did not necessarily have expert knowledge about the product, which may have helped them to think "outside the box". Finally, they were not inhibited by listening to others blocking thinking or the fear of being criticized by others, which may have led to longer lists of ideas generated by individuals.

During the group-brainstorming sessions, people from different disciplines (marketing, R&D, supply chain) were present. This may have led to early evaluations, in which ideas that had been looked into in the past and ideas that seemed commercially unviable were not included. This could potentially have reduced the number of radical ideas generated. Nonetheless, Sutton and Hargadon (1996) described group brainstorming positively, because of its potential to generate profitable ideas supporting the organizational memory. In this case the ideas generated by the groups may be the ones that best meet current consumer needs. These options however are unlikely to be the ones that enable the best improvement in greenhouse gas emissions.

The aim of the eco-ideation tool developed in this research is to identify eco-innovative options for one environmental impact theme: greenhouse gas emission reductions. This makes sense for products for which greenhouse gases are the main environmental burden (as opposed to other metrics such as water and waste). The tool could be expanded to include other environmental impacts such as waste for which the link between design and impacts is widely understood (this is for instance more difficult for toxicity). However, the tool now has a clear focus and users do not need to "weigh" different environmental burdens. It is important however to be aware of other environmental impacts when generating new ideas.

Having developed a tool for idea generation, the authors now plan to develop a simplified assessment of the ideas, to contrast the 'gain' (emissions reduction) from each idea against the 'pain' (commercial difficulty) of implementation. In addition, because group-brainstorming sessions are popular, the use of the tool in bigger groups is explored. During two separate initial group trials using the tool, over 60 and 100 ideas respectively were generated in only 2 h. Each group consisted of 5 people and one author served as a facilitator. The aim of the group session during which 60 ideas were generated was to generate ideas about more *durable* products (refrigeration systems), as opposed to the fast moving consumer goods for which the tool was initially developed. The eco-ideation tool thus can potentially be applied to wider contexts, although the questions may need to be rephrased slightly. More trials are required to confirm the apparent benefit of using this tool, and the tool will also be tested in other sectors.

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